

Program4: Integrate git with docker for source controlled application build

Step1: create a directory and install npm

mkdir program4

cd program4

To install npm

>> npm init -y

```
rv24mc053_kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa1xxx:~$ mkdir program4
rv24mc053_kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa1xxx:~$ cd program4
rv24mc053_kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa1xxx:~/program4$ npm init -y
Wrote to /home/rv24mc053_kavya/program4/package.json:

{
  "name": "program4",
  "version": "1.0.0",
  "description": "",
  "main": "index.js",
  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1"
  },
  "keywords": [],
  "author": "",
  "license": "ISC"
}
```

Step2: create index.js and install express(npm install express)

>> nano index.js

```
rv24mc053_kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa1xxx:~/program4$ nano index.js
rv24mc053_kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa1xxx:~/program4$ npm install express

added 68 packages, and audited 69 packages in 2s

18 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities
rv24mc053_kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa1xxx:~/program4$ nano Dockerfile
```

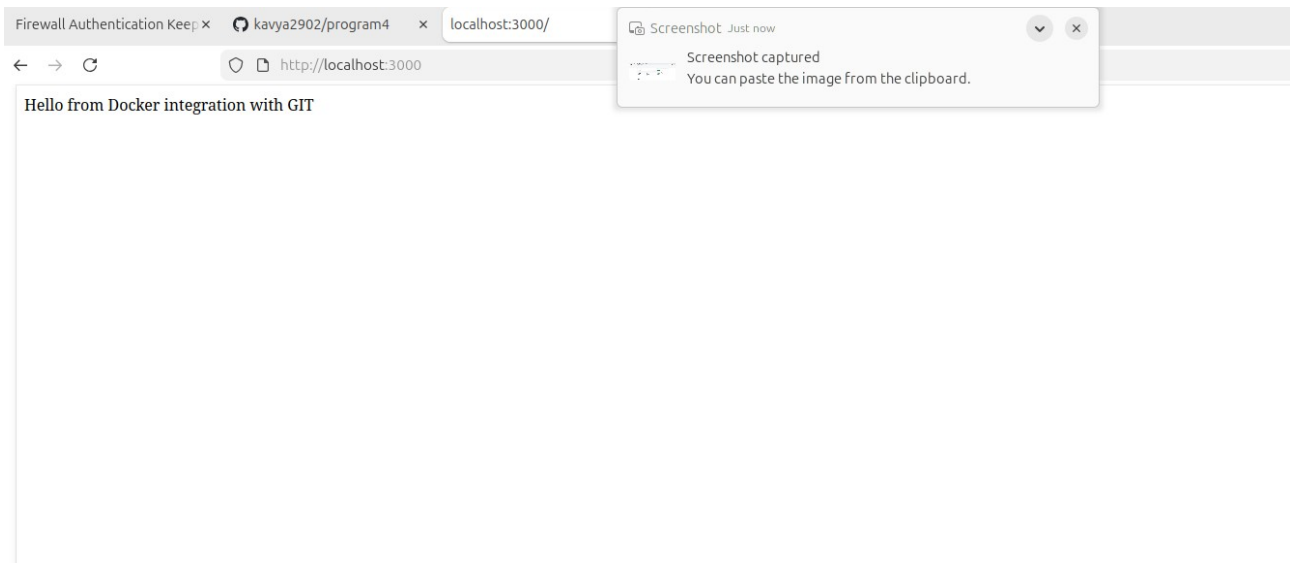
Step3: create a repository in github and and commands to push the source code

```
rv24mc053_kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa1xxx:~/program4$ git init
Initialized empty Git repository in /home/rv24mc053_kavya/program4/.git/
rv24mc053_kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa1xxx:~/program4$ git add .
rv24mc053_kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa1xxx:~/program4$ git config --global user.name "kavya2902"
rv24mc053_kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa1xxx:~/program4$ git config --global user.email "kr4273020@gmail.com"
rv24mc053_kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa1xxx:~/program4$ git commit -m "Initial commit"
[master (root-commit) e3d549d] Initial commit
636 files changed, 72757 insertions(+)
create mode 100644 Dockerfile
create mode 100644 index.js
create mode 100644 node_modules/.package-lock.json
create mode 100644 node_modules/accepts/HISTORY.md
create mode 100644 node_modules/accepts/LICENSE
create mode 100644 node_modules/accepts/README.md
create mode 100644 node_modules/accepts/index.js
create mode 100644 node_modules/accepts/package.json
create mode 100644 node_modules/body-parser/HISTORY.md
create mode 100644 node_modules/body-parser/LICENSE
create mode 100644 node_modules/wrappy/README.md
create mode 100644 node_modules/wrappy/package.json
create mode 100644 node_modules/wrappy/wrappy.js
create mode 100644 package-lock.json
create mode 100644 package.json
rv24mc053_kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa1xxx:~/program4$ git branch -M main
rv24mc053_kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa1xxx:~/program4$ git remote add origin https://github.com/kavya2902/program4.git
rv24mc053_kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa1xxx:~/program4$ git push -u main
fatal: 'main' does not appear to be a git repository
fatal: Could not read from remote repository.

Please make sure you have the correct access rights
and the repository exists.
rv24mc053_kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa1xxx:~/program4$ git push -u origin main
Username for 'https://github.com': kavya2902
Password for 'https://kavya2902@github.com':
Enumerating objects: 719, done.
Counting objects: 100% (719/719), done.
Delta compression using up to 12 threads
Compressing objects: 100% (666/666), done.
Writing objects: 100% (719/719), 725.06 KiB | 3.98 MiB/s, done.
Total 719 (delta 143), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (143/143), done.
To https://github.com/kavya2902/program4.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
```

Step4: Build the Dockerfile and to see the output enter the url in the browser

```
rv24mc053_kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa1xxx:~/program4$ docker build -t my_node_app .
[+] Building 7.8s (14/14) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 255B
=> [internal] load metadata for docker.io/library/node:18-slim
=> [internal] load metadata for docker.io/library/node:18
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [internal] load build context
=> => transferring context: 3.76MB
=> [stage-1 1/3] FROM docker.io/library/node:18-slim@sha256:f9ab18e354e6855ae56ef2b290dd225c1e51a564f87584b9bd21dd651838830e
=> [build 1/5] FROM docker.io/library/node:18@sha256:c6ae79e38498325db67193d391e6ec1d224d96c693a8a4d943498556716d3783
=> CACHED [build 2/5] WORKDIR /app
=> CACHED [build 3/5] COPY package*.json ./
=> CACHED [build 4/5] RUN npm install
=> [build 5/5] COPY . .
=> CACHED [stage-1 2/3] WORKDIR /app
=> [stage-1 3/3] COPY --from=build /app /app
=> exporting to image
=> => exporting layers
=> => writing image sha256:c958103d02521261c0b7f07a64a379b33e5caf0d7b285e50364ef087d8406223
=> => naming to docker.io/library/my_node_app
rv24mc053_kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa1xxx:~/program4$ docker run -p 3000:3000 my_node_app
Server running on http://localhost:3000
```



Step5: check whether the source code is pushed to git or not
-- → open the repo and refresh it now u can the source code as given in the below image

